

A3. Static phase converters are used for operating three-phase equipment with single-phase power supply. With the help of a three-phase generating rotary phase converter, you can operate three-phase equipment from normal residential single-phase power supply.

Static phase converters operate by charging and discharging capacitors to temporarily produce a third phase of power for only a matter of seconds during the startup of electric motors. It then drops out forcing the motor to continue to run on just one phase.

Static phase converters do not actually convert single-phase power into three-phase power, nor do these generate three-phase power. Due to their technology, they do not properly power a three-phase machinery or equipment. They have limited utility as they are unable to power all types of machinery with three-phase circuitry. They are also low on the reliability scale.

The performance of static phase converters is inferior to a true rotary phase converter built with a three-phase generator that provides true three-phase power. Heavy-duty, rotary phase converters generate high-quality, three-phase power while running on a single phase. They use a balanced

three-phase generator with power factor correction to provide balanced, safe three-phase power.

Such three-phase converters may allow the frequency to be varied, allowing for different equipment frequency requirements (50 Hz, 60 Hz, 400 Hz, etc) and also for motor-speed control (variable-frequency drives (VFDs)).

Other types of converters include phase-inverter VFDs and CNC rotary phase converters. A CNC phase converter is simply a rotary phase converter with balanced three-phase output to provide three-phase power to sensitive computers that are part of CNC machines.

AC-to-DC, DC-to-AC or DC-to-DC are also converters but of different types. They only change the format and level of the voltage at the input terminals. Power supplies, UPS and inverters fall under this category.

Q4. What is the difference between electrostatic discharge grounding and equipment grounding?

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A4. *Grounding for protection against electrostatic discharge.* Static electricity is generated during machine operation of an equipment or material handling by the operating personnel. The level of charge is affected by the material type, speed of contact and separation, humidity and several other factors. Discharge of static electricity can damage devices or result in latent damage that may skip immediate attention but cause the device to fail prematurely once in service.

Electrostatic discharge can occur throughout the manufacturing, test, shipping, handling or operational processes. Therefore effective electrostatic discharge control is required for controlling the damage.

Design of the equipment with immunity to the effects of

electrostatic discharge, its elimination and reduction of generation, and effective dissipation or neutralisation form part of electrostatic discharge control.

A primary means of protecting ESD-susceptible items is to provide a ground path to bring ESD-protective materials and personnel to the same electrical potential by bonding or electrically connecting to a known electrical ground point, creating an equipotential balance between all items and personnel.

As the overall strategy to protect the system against generation and dissipation of static electricity has many facets other than electrostatic discharge grounding, you must refer to electrostatic discharge standards as well.

Equipment grounding. For grounding of electrical equipment, conductive materials enclosing electrical conductors or equipment, or part of such equipment, are connected to the earth to limit the voltage to ground on these materials. The grounding provides a path and these parts energise as a result of the failure of the insulation of a conductor inside the equipment. It is the primary way to protect personnel from electrocution. It also provides an intentional path of ample current-carrying capacity and low impedance to insure rapid operation of the circuit's over-current protection under ground fault conditions.

In electronics, different grounds may exist. Electromagnetic, electrostatic, signal, chassis and power grounds are addressed separately in order to control noise, interference and undesirable current flow.

For more information on issues related to grounding, you should refer to sources that provide information, guidelines and standards for effective grounding, aspects of electrical safety in plants and machinery, and protection against electrical shock hazard.

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A typical rotary phase converter (Model PAM-RC-5 (5 HP))